Signals and Systems

Week 3: Fourier series

Lecture slides for EE2001

Why Fourier series?

- Any periodic signal is a sum of sinusoids
- Each harmonic has its own amplitude and phase
- A filter acts on each harmonic separately

Worked example

A square wave at 50 Hz has only odd harmonics.

- Third harmonic: 150 Hz
- Fifth harmonic: 250 Hz
- Seventh harmonic: 350 Hz

Summary

Periodic signals can be analysed one harmonic at a time.

Next week: the Fourier transform for signals that do not repeat.